

Learning How to Learn: What the Numbers Hide

学会学习：数字隐藏了什么

Theme: 学习方法与教育 · New: 45 · Review: 18 · Planned cumulative progress: 45/6, 127

English Story

The charge came **early** and it was **explicit**: the group had measured the wrong thing. A visiting examiner glanced at their table and asked what the numbers **could mean**. “You count arrivals,” she said. “You cannot see **comprehension**.” There was a **lack** of **wisdom** in the **early** draft, and they knew it. The criticism was not **commercial**; it was about **territory**. **But** their numbers did not **increase**, and that was the first sign of trouble.

A **sensible** answer **would** separate **retention** from recognition. A student who can **recite** a definition may still fail to **recognize** the same idea in a **new** context. **Memory**, they wrote, is not a **certificate**; it is a **presence** that decays unless it is used. The **decrease** was **visible** in their own data, and the **meaning** of that **line** was the **catch**.

The **obstacle** was **perception**. A skilled reader **could penetrate** a **vague** sentence at a glance, and a beginner **could** not; an **oval** diagram of their results made that difference **noticeable**, and her **finger** followed it while she talked. “You should **instruct** with an **illustrative** example,” Professor Zhao said. “Show the **glimpse** of understanding, not the **purchase** of a book.”

They began to **oppose** the easy number. A **gifted** singer, a **rental** clerk, and a boy with a **rifle** on a poster all appeared in the same week’s notes, and none of them **could qualify** as evidence. “We must **greet** the sceptic,” Maya said, “and not be bothered by her.” It was a **sophisticated** position, and it took **sophistication** to defend it without **sparkle**. **Other** volunteers agreed; a **kid** from the first year asked the sharpest question of the evening.

So they rebuilt the study around two questions: what **would** a reader remember, and what **could** a reader do? An **invention** of their own — a seven-**line** test — became the **new** instrument, and its **performance** improved the whole file. They were **readily aware** of its limits, and they did not **regret** stating them. “Do not let the outside world **intrude** on the **meaning**,” Lin wrote. “**Enlighten** it instead.” **Later**, a **fresh teacher** asked whether the test prevented guessing; the honest answer was that it did not, and they wrote that down too. “**Yeah**,” Maya said. “**Turn** the page and start again.”

中文翻译

指控来得很早，而且明确：小组测量错了东西。一位来访的考官瞥了一眼他们的表格，问这些数字能意味着什么。“你们数的是到场人数，”她说，“你们看不到理解。”早期草稿里缺少智慧，他们自己也知道。这项批评不是商业性的，而是地盘性的。但他们的数字没有增加，而这是麻烦的第一个信号。

一个明智的答案会把保持与识别分开。一个能背出定义的学生，仍可能在新情境中认不出同一个概念。他们写道：记忆不是一张证书，而是一种会衰败的在场，除非被使用。下降在自己数据里是可见的，而那一行的含义正是陷阱所在。

障碍在于知觉。熟练的读者一瞥就能穿透含糊的句子，初学者却不能；他们结果的那张椭圆图把这种差别显现得相当明显，她说话时手指一直顺着它移动。“你应该用一个作例证的例子来讲，”赵教授说，“展示理解的一瞥，而不是购买一本书。”

他们开始反对那个容易的数字。一位有天赋的歌手、一位租车店员，以及海报上一个拿着步枪的男孩都出现在同一周的笔记里，而他们当中没有一个能算作证据。“我们必须欢迎怀疑者，”曼雅说，“而不是被她烦到。”这是一个久经世故的立场，而要既捍卫它又不靠卖弄光彩，需要同样的世故。其他志愿者也同意了；一个一年级的孩子提出了当晚最尖锐的问题。

于是他们围绕两个问题重建研究：读者会记住什么，读者能做什么？他们自己的一项发明——一份七行测试——成了新工具，它的表现改善了整个档案。他们很快就意识到它的限度，也并不后悔把它写出来。“不要让外部世界闯入意义，”林伟写道，“而是去启发它。”后来，一位新来的教师问这项测试是否防止了猜测；诚实的回答是并没有，他们把这一点也写了下来。“是的，”曼雅说，“翻过这一页，重新开始。”

Core Vocabulary · 核心词汇

Word / Pronunciation	Context meaning / source example
would wud POS: aux	aux. 将, 愿意 Collocation: — (not independently sourced) Example: imp. of Will; v. t. Commonly used as an auxiliary verb, either in the past; tense or in the conditional or o...
could kud POS: aux	aux. 可以, 能 Collocation: — (not independently sourced) Example: imp. of Can; imp. Was, should be, or would be, able, capable, or; susceptible. Used as an auxiliary, in the...
mean mi:n POS: a	a. 低劣的, 卑贱的, 简陋的, 吝啬的, 惭愧的, 平均的, 中间的, 普通的; vt. 意谓, 想要, 意欲, 预定; vi. 用意, ... Collocation: — (not independently sourced) Example: n. an average of n numbers computed by adding some function of the numbers and dividing by some function of n...
increase in'kri:s POS: n	n. 增加, 增进, 利益; vt. 增加, 加大; vi. 增加, 繁殖 Collocation: — (not independently sourced) Example: n. a change resulting in an increase; n. a process of becoming larger or longer or more numerous or more impor...
recognize 'rekegnaiz POS: vt	vt. 认出, 认可, 承认, 公认, 识别, 赏识; vi. 承认, 具结 Collocation: — (not independently sourced) Example: v. be fully aware or cognizant of; v. perceive to be the same; v. exhibit recognition for (an antigen or a subs...
memory 'meməri POS: n	n. 记忆, 记忆力, 回忆, 纪念, 存储; n. 内存; [计] 存储器, 内存, 查看内存实用程序 Collocation: — (not independently sourced) Example: n. something that is remembered; n. the cognitive processes whereby past experience is remembered; n. the power...
aware ə'weə POS: a	a. 知道的, 有觉悟的 Collocation: — (not independently sourced) Example: a. (sometimes followed by 'of') having or showing knowledge or understanding or realization or perception
presence 'preznz POS: n	n. 出席, 面前, 存在, 仪态, 风度; [电] 出现 Collocation: — (not independently sourced) Example: n. the state of being present; current existence; n. the immediate proximity of someone or something; n. an inv...
meaning 'mi:nɪŋ POS: n	n. 意义, 含义, 目的, 意图; a. 意味深长的 Collocation: — (not independently sourced) Example: n. the message that is intended or expressed or signified; n. the idea that is intended; s. rich in significanc...
perception pe'seɪpʃən POS: n	n. 知觉, 感觉, 领悟力, 获取; [医] 知觉 Collocation: — (not independently sourced) Example: n. a way of conceiving something; n. the process of perceiving; n. knowledge gained by perceiving
oppose ə'pəuz POS: vt	vt. 反对, 以...对抗, 抗争; vi. 反对 Collocation: — (not independently sourced) Example: v. be against; express opposition to; v. contrast with equal weight or force; v. be resistant to
territory 'terɪtəri POS: n	n. 领土, 领地, 版图, 地区, 活动范围; [经] (推销员等的) 推销区域 Collocation: — (not independently sourced) Example: n. an area of knowledge or interest; n. the geographical area under the jurisdiction of a sovereign state
bother 'bɒðə POS: vt	vt. 烦扰, 迷惑; vi. 烦恼, 操心; n. 麻烦, 纠纷, 讨厌的人 Collocation: — (not independently sourced) Example: v. intrude or enter uninvited; v. make nervous or agitated; v. make confused or perplexed or puzzled
purchase 'pɜ:tʃəs POS: n	n. 购买, 购买品, 紧握, 绞辘; vt. 购买, 赢得, 努力取得, 用滑轮起(锚等) Collocation: — (not independently sourced) Example: n. the acquisition of something for payment; n. something acquired by purchase; n. a means of exerting influenc...
visible 'vɪzəbl POS: a	a. 看得见的, 明显的, 显然的; n. 可见物 Collocation: — (not independently sourced) Example: a. capable of being seen; or open to easy view; s. obvious to the eye; s. present and easily available
certificate sə'tɪfɪkeɪt POS: n	n. 证书, 证明书; vt. 发给证明书, 用证书批准, 用证书证明 Collocation: — (not independently sourced) Example: n. a document attesting to the truth of certain stated facts; v. present someone with a certificate; v. authori...
qualify 'kwɒlɪfaɪ POS: vi	vi. 取得资格, 有资格; vt. 使有资格, 使合格, 限定, 限制, 准予 Collocation: — (not independently sourced) Example: v. prove capable or fit; meet requirements; v. pronounce fit or able; v. make more specific; v. make fit or prep...
sensible 'sensəbl POS: a	a. 有感觉的, 敏感的, 明智的; [医] 可感觉的 Collocation: — (not independently sourced) Example: a. able to feel or perceive; s. readily perceived by the senses; s. aware intuitively or intellectually of some...
readily 'redɪli POS: adv	adv. 迅速地, 轻易地, 乐意地 Collocation: — (not independently sourced) Example: r. without much difficulty
wisdom 'wɪzdəm POS: n	n. 智慧, 明智行为, 学识, 名言, 贤人 Collocation: — (not independently sourced) Example: n. accumulated knowledge or erudition or enlightenment; n. the trait of utilizing knowledge and experience wit...

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sophisticated sə'fistiketɪd POS: a	a. 复杂的, 久经世故的; [法] 尖端的, 高级的, 非常有经验的 Collocation: — (not independently sourced) Example: a. having or appealing to those having worldly knowledge and refinement and savoir-faire; s. intellectually ap...
rifle 'raɪfl POS: n	n. 来复枪, 步枪; vt. 用步枪射击, 抢夺; vi. (抱有不良意图) 迅速翻查 Collocation: — (not independently sourced) Example: n. a shoulder firearm with a long barrel and a rifled bore; v. go through in search of something; search throu...
gifted 'giftɪd POS: a	a. 有天才的 Collocation: — (not independently sourced) Example: v give qualities or abilities to; v give as a present; make a gift of; a endowed with talent or talents
greet gri:t POS: vt	vt. 问候, 致敬, 欢迎, 映入眼帘 Collocation: — (not independently sourced) Example: v. express greetings upon meeting someone; v. send greetings to; v. react to in a certain way; v. be perceived by
decrease 'di:kri:s POS: n	n. 减少, 减少量; v. 减少 Collocation: — (not independently sourced) Example: n. a change downward; n. a process of becoming smaller or shorter; n. the amount by which something decreases; n...
obstacle 'ɒbstəkl POS: n	n. 障碍, 妨害物, 阻碍 Collocation: — (not independently sourced) Example: n. something immaterial that stands in the way and must be circumvented or surmounted; n. an obstruction that...
explicit ɪk'splɪt POS: a	a. 详述的, 清楚的, 直言的 Collocation: — (not independently sourced) Example: a. precisely and clearly expressed or readily observable; leaving nothing to implication
regret rɪ'ɡret POS: n	n. 遗憾, 后悔, 悔恨, 抱歉, 歉意; vt. 为...感到遗憾, 后悔, 惋惜, 懊悔, 抱歉; vi. 感到抱歉 Collocation: — (not independently sourced) Example: v. feel sad about the loss or absence of; v. decline formally or politely; v. express with regret
instruct ɪn'strʌkt POS: vt	vt. 教, 教育, 命令, 通知; [法] 托办, 指导, 指示 Collocation: — (not independently sourced) Example: v. give instructions or directions for some task; v. make aware of
vague veɪɡ POS: a	a. 含糊的, 不清楚的, 茫然的 Collocation: — (not independently sourced) Example: s not clearly understood or expressed; a not precisely limited, determined, or distinguished; s lacking clarity...
glimpse ɡlɪmps POS: n	n. 一瞥, 一闪; vi. 投以一瞥, 闪烁不定; vt. 瞥见 Collocation: — (not independently sourced) Example: n. a brief or incomplete view; n. a vague indication; v. catch a glimpse of or see briefly
rental 'rentl POS: n	n. 租费, 租金收入; a. 租赁的, 收取租金的 Collocation: — (not independently sourced) Example: n. the act of paying for the use of something (as an apartment or house or car); a. available to rent or lease...
invention ɪn'venʃən POS: n	n. 发明, 创作能力, 虚构的故事; [经] 发明 Collocation: — (not independently sourced) Example: n. the creation of something in the mind; n. a creation (a new device or process) resulting from study and exp...
penetrate 'penɪtreɪt POS: vt	vt. 穿透, 刺穿, 渗透, 看穿, 洞察, 了解, 使充溢; vi. 刺入, 看穿, 渗透, 打进 Collocation: — (not independently sourced) Example: v. pass into or through, often by overcoming resistance; v. come to understand; v. make one's way deeper into o...
decay di'keɪ POS: n	n. 衰退, 腐败; v. (使)衰退, (使)腐败 Collocation: — (not independently sourced) Example: n. the process of gradually becoming inferior; n. a gradual decrease; as of stored charge or current; n. the or...
noticeable 'nəʊtɪsəbl POS: a	a. 显而易见的, 显著的, 值得注意的 Collocation: — (not independently sourced) Example: a. capable or worthy of being perceived; s. readily noticed
recite rɪ'saɪt POS: v	v. 背诵, 朗读, 叙述 Collocation: — (not independently sourced) Example: v. repeat aloud from memory; v. render verbally, "recite a poem"
retention rɪ'tenʃən POS: n	n. 保持, 留置, 保留, 保持力, 保留物, 记忆, 停滞, 固位; [计] 保留, 保持 Collocation: — (not independently sourced) Example: n. the act of retaining something
oval 'əʊvəl POS: n	n. 椭圆形; a. 卵形的, 椭圆形的 Collocation: — (not independently sourced) Example: n a closed plane curve resulting from the intersection of a circular cone and a plane cutting completely thro...
comprehension ,kɒmpri'henʃən POS: n	n. 理解, 包含 Collocation: — (not independently sourced) Example: n. an ability to understand the meaning or importance of something (or the knowledge acquired as a result)
sophistication sə'fɪstɪ'keɪʃən POS: n	n. 诡辩, 老于世故, 复杂(性); [医] 掺假 Collocation: — (not independently sourced) Example: n. being expert or having knowledge of some technical subject; n. the quality or character of being intellectu...

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sparkle 'spa:kl POS: n	n. 闪耀, 火花, 活力, 发泡; vi. 闪耀, 冒火花; vt. 使闪耀 Collocation: — (not independently sourced) Example: n. merriment expressed by a brightness or gleam or animation of countenance; v. reflect brightly; v. be lively...
intrude in'tru:d POS: vi	vi. 闯入, 侵入; vt. 强加于 Collocation: — (not independently sourced) Example: v. enter uninvited; v. search or inquire in a meddlesome way; v. thrust oneself in as if by force
enlighten in'laitn POS: vt	vt. 教育, 启发, 启蒙, 开导, 教导 Collocation: — (not independently sourced) Example: v. make understand; v. give spiritual insight to; in religion
illustrative 'ilestrativ POS: a	a. 说明的, 解说的, 作例证的 Collocation: — (not independently sourced) Example: s clarifying by use of examples; s serving to demonstrate